Atlas-Based Image Reconstruction for Breast CT Imaging Using Non-Isocentric C-Arm Scanner

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***Abstract*—Cone-beam computed tomography (CBCT) is known to provide high-quality three-dimensional (3D) images for accurate diagnosis of breast cancer compared to conventional two-dimensional (2D) mammography. Several CBCT techniques have been developed for breast imaging with different imaging setup and geometry configurations. In interventional procedures such as biopsy, it is likely that a 3D image with depth information is required. This motivates us to study the use of a generic purpose non-isocentric C-arm scanner in 3D breast imaging. The imaging equipment used in this study is known to have several design configurations so that its use in 3D tomographic imaging is limited. Recently, we have developed an image reconstruction method, named PA-MAP algorithm, that incorporates prior information obtained from probabilistic atlas to improve the image quality in low-dose CT. The atlas can be generated from a set of earlier scans of different patients to provide a prior information in the form of probabilistic atlas. It is proved that atlas prior can significantly reduce image artifacts in several imaging scenarios with limited data. In this paper, we study the use of PA-MAP algorithm in CBCT exploiting non-isocentric C-arm scanner for breast imaging.**

I. Introduction

The development of breast CT imaging systems is an active development track with several prototypes presented to the community [1–4]. Cone-beam computed tomography (CBCT) is known to provide a high-quality 3D images for accurate diagnosis of breast abnormalities compared to conventional 2D mammography [5]. Most of breast CT imaging systems are based on flat-panel detectors and circular trajectory with analytical image reconstruction techniques. Clinical interventional procedures such as breast biopsy are normally operated under the guide of C-arm scanner. It is demonstrated that breast tomosynthesis can be successfully achieved using C-arm scanner [6]. However, these devices can only provide 2D images with no depth information. To obtain 3D image, a patient is required to move to another room for tomographic imaging using x-ray CT. It would be more helpful to use the same device to generate tomographic image for patient's convenient and saving procedure time. This motivates us to study the use of a generic purpose non-isocentric C-arm scanner in 3D breast imaging. The imaging equipment used in this study is known to have several design configurations that limit its use in 3D tomographic imaging. The C-arm orbital movement is normally limited to angular range around $\frac{2\pi}{3}$, which is far below the theoretically required for exact reconstruction. Also, the non-isocentric design is known to limit the potential use in 3D imaging as the scanner field-of-view (FOV) is non-stationary and is view-angle dependent. Moreover, it is difficult to acquire enough number of projection views with exact angulation value through a manually operated C-arm scanner. The challenges associated with non-isocentric C-arm geometry in comparison to isocentric ones are demonstrated in Fig. 1.

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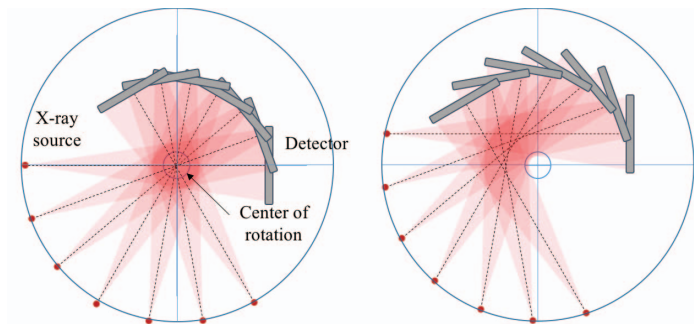

Fig. 1. Planner sketch of C-arm gantry positions with isocentric (left) and non-isocentric (right) geometries. Few projection views over limited orbital range are shown to demonstrate the geometry limitations in non-isocentric setup.

Recently, we have developed an image reconstruction method, named PA-MAP algorithm, that incorporates prior information obtained from a probabilistic atlas to improve image quality in low-dose CT [7]. The atlas can be generated from a set of earlier scans of different patients. In this paper, we study the use of PA-MAP algorithm in CBCT exploiting non-isocentric C-arm scanner for breast imaging.

II. Methods

A. Extended trajectory

Most of mobile non-isocentric C-arm scanners are provided with limited orbital rotation range, which is convienient for interventional procedures or surgical planning [8]. We solve this problem by using an extended trajectory setup. The proposed trajectory is implemented through two steps. In the first step, we rotate the C-arm around the target object as shown in Fig. 2(a). The acquired data will cover an orbital range of about $\frac{2\pi}{3}$ in most commercial devices. The second step is implemented by moving up the C-arm elevator, then rotate the whole C-arm with angulation angle of $180°$ as shown in Fig. 2(b). The elevator movement value can be easily computed by the scanner configurations. The proposed

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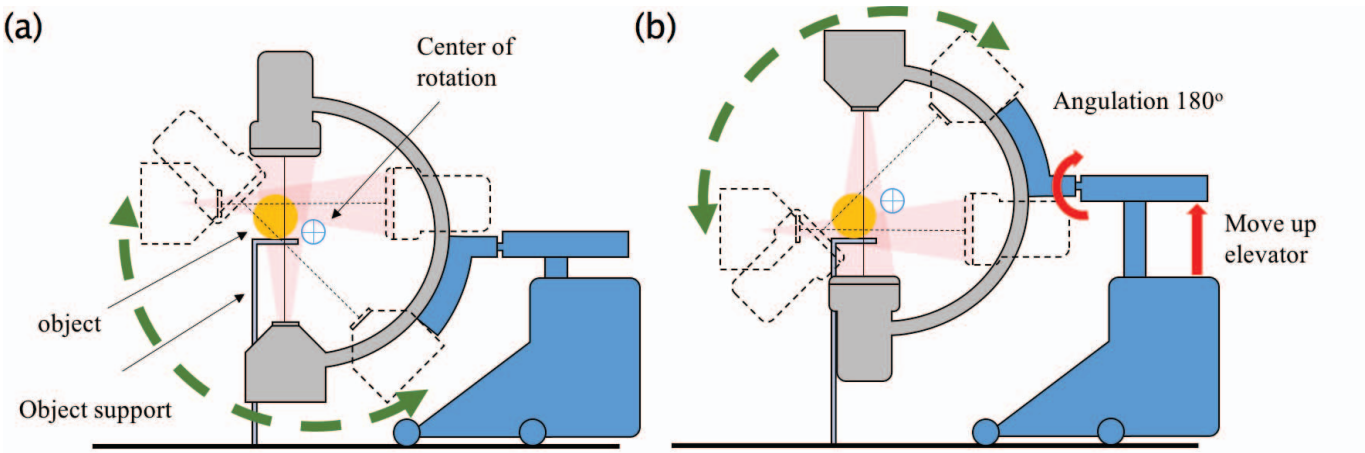

Fig. 2. Extended C-arm trajectory using flip-up strategy. (a) First step considering object scanning over a complete C-arm orbital rotation. (b) Second step includes a flip-up positing followed by a complete C-arm rotation. Scanned object is stationary and fixed using object support devise.

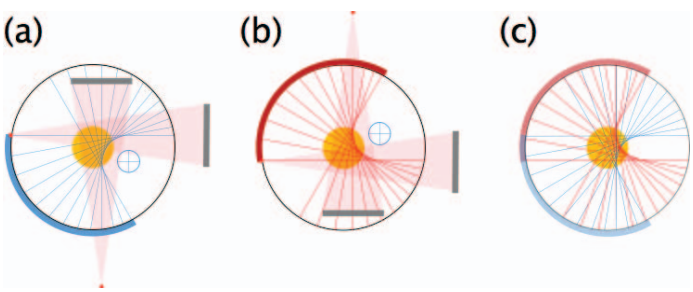

Fig. 3. Extended trajectory for non-isocentric C-arm scanner. (a) Initial trajectory acquired from complete C-arm rotation. Blue lines indicate the center line between x-ray source and detector. Blue arc defines the angular range of the x-ray source. The position of imaging system in horizontal and vertical directions is also shown. (b) The additional trajectory through flip-up position. (c) The extended trajectory which generated through a combination of trajectories in (a) and (b).

extended trajectory can result in an orbital rotation that is required for theoretically exact reconstruction (*i.e.* 180° + fan angle) as demonstrated in Fig. 3. The C-arm rotation and positioning is implemented by using the system we have developed earlier [9]. This system consists of a gantry control unit (GCU) which is a motor mounted to the C-arm and an inertial measurement unit (IMU) that is used to acquire the current position geometries. The single C-arm rotation requires about 20 seconds and the projection data is acquired as a single cine loop. The completed procedure time is around two minutes as the angulation used to alter the C-arm position is currently performed manual.

B. Probabilistic atlas construction

In this section, we briefly describe how to construct the atlas prior [7]. The Laplacian mixture model (LMM) is used to decompose the sample data into different clusters. The density function at an observation $\mathbf{x} = (x_1, \ldots, x_n)$ is defined as:

$$p(\mathbf{x}) = \prod_{j=1}^{n} \sum_{l=1}^{L} \pi_{lj} p(x_j|\Omega_l), \tag{1}$$

where $\Omega_l (l = 1, \ldots, L)$ is the class labels and π_{jl} is the prior probability for each image pixel x_j assigned to mixture component Ω_l. The probability density function $p(x_j|\Omega_l)$ is defined as:

$$p(x_j|\Omega_l) = \frac{1}{2\lambda_l} \exp\left(-\frac{|x_j - \mu_l|}{\lambda_l}\right), \tag{2}$$

where μ_l and λ_l are the median and width parameter of the density function corresponding to the label Ω_l. Considering the LMM in Eq. (1), the atlas is constructed by using a dataset of pre-registered scans of normal cases, which is used to generate an atlas representing normal tissues. The atlas information is then embedded into the image reconstruction as described below.

C. Image reconstruction

The imaging geometry presented here is challenging and consists of several data limitations (limited-views, limited-angle and offset-scan), which are known to degrade the quality of the reconstructed images. The use of prior information in terms of statistical reconstruction framework is known to be helpful in decreasing image artifacts. Consider that the detector measurements $\mathbf{y} = (y_1, \ldots, y_m)$ are acquired with $m \times n$ system matrix $A = \{a_{ij}\}$ and blank scan measurements $\mathbf{b} = (b_1, \ldots, b_m)$, the statistical model that represents the tomographic imaging is:

$$y_i \approx \text{Poisson}[b_i \exp(-\langle a_i, \mathbf{x}\rangle)], \quad i = 1, \ldots, m. \tag{3}$$

The proposed image reconstruction cost function is defined as:

$$\min_{\mathbf{x} \geq 0} f(\mathbf{x}) = L(\mathbf{x}) + \alpha D(\mathbf{x}) + \beta U(\mathbf{x}), \tag{4}$$

$$L(\mathbf{x}) = \sum_{i=1}^{m} [b_i \exp(-\langle a_i, \mathbf{x}\rangle) + y_i \langle a_i, \mathbf{x}\rangle], \tag{5}$$

$$D(\mathbf{x}) = -\sum_{j=1}^{n} \log\left[\sum_{l=1}^{L} \pi_{lj} p(x_j|\Omega_l)\right], \tag{6}$$

$$U(\mathbf{x}) = \sum_j \sum_{j' \in N_j} \omega_{jj'} \psi(t), \quad \psi(t) = \log(1 + |\frac{t}{\delta}|), \qquad (7)$$

where $L(\mathbf{x})$ is the negative log-likelihood function, $D(\mathbf{x})$ is the atlas prior function and $U(\mathbf{x})$ is a smoothing penalty. The hyperparameters α and β are used to control the strength of the priors.

III. Results and Discussion

A simple 3D breast phantom is generated with size ranged from 166 to 234 mm (from 128 to 180 mm) in horizontal (vertical) direction. We generate 150 virtual objects with size variations. These objects are used to construct a simple atlas which consists of three components (air, low-density tissues and high-density tissues), i.e. $L = 3$. A single object with size of 200 mm and 154 mm is used as the target object to be reconstructed. A set of abnormalities are added to the phantom as shown in Fig. 4(a). We consider four large-size abnormalities and an array of 16 small size abnormalities (microcalcifications). We simulate a real scanner geometry parameters associated to the clinical C-arm scanner (Ziehm Vision R, Ziehm Imaging GmbH, Nürnberg, Germany). The source to detector distance is 970 mm, detector to center-of-rotation is 364 mm, detector offset is 130 mm, detector size is 310 mm and image grid is 512×512 pixels. The position of the phantom is estimated by the ray-count method detailed in [10]. In the first experiment, the projection data are acquired using 32, 64 and 128 views over orbital range of 115°. We consider the single rotation scheme (shown in Fig. 2(a) only). The reconstruction is achieved through 150 iterations of the OS-Convex algorithm (2 subsets) [11] and the PA-MAP algorithm (2 subsets and $\beta = 0.1$). We used dynamic values of parameter $\alpha = \alpha_0/(k+1)$, where k is the iteration number and $\alpha_0 = 100$. The comparison with OS-Convex algorithm is used as the data fidelity term of the PA-MAP algorithm is implemented through the same method [7]. From the reconstruction results shown in Fig. 5, it is observed that the PA-MAP algorithm is able to preserve both large size abnormalities in terms of shape structure and small size abnormalities in terms of contrast. Even with the challenging single-pixel microcalcifications, most of the pattern is preserved with good diagnostic quality compared to conventional method. However, we still observe a deformation in the object general structure in both methods due to the limited orbital range.

In the second experiment, we repeated the above step, but we consider the extended trajectory using geometries shown in Fig. 2(a) and (b). We consider 8+8, 16+16 and 32+32 projection views. Reconstruction results are shown in Fig. 6. Although the number of projection views is rather smaller than those used in the first experiment, we can observe enhancement in the shape deformation as the external object contour can be recovered correctly. Also, the quality of large size abnormalities is improved in the PA-MAP images.

We conduct additional experiment using real-data obtained from chest CT. A single breast slice is extracted and artificial calcification-like abnormalities are added as shown in Fig. 7(a). The same atlas was registered to the breast slice using non-rigid registration and affine transformation. Reconstruction from 16+16, 32+32 and 48+48 projection views are implemented using 50 iterations with $\beta = 0.1$ and $\alpha_0 = 10$. Results are shown in Fig. 7. In OS-Convex reconstruction, streak artifacts are quite strong and it is rather difficult to observe anatomical structure of the breast. In contrast, PA-MAP images looks very close to the true image with image quality improvements in terms of both noise properties and contrast.

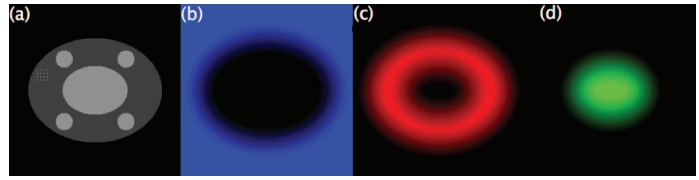

Fig. 4. (a) Breast phantom with added abnormalities. (b) - (d) the probabilistic atlas representing air, low-density tissues and high-density tissues, respectively. This atlas is constructed from 150 samples of the breast phantom (without abnormalities) generated using different sizes.

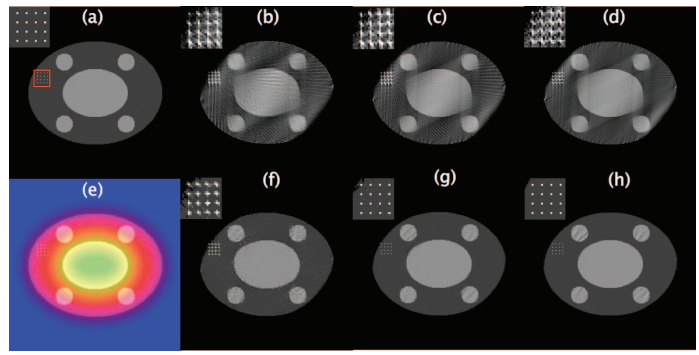

Fig. 5. (a) True object with magnification of microcalcifications marked with red square shown in the top left corner. (e) is a merge image of true object registered with the probabilistic atlas. (b)-(d) are the OS-Convex reconstruction using 32, 64 and 128 projection views, respectively. (f)-(h) are the PA-MAP reconstruction using 32, 64 and 128 projection views, respectively. The magnification of the same region shown in (a) is also generated for each reconstruction for comparison.

IV. Conclusion

In this paper, we investigate a new imaging setup for breast CT. The goal is to provide a 3D scan of the breast using general purpose C-arm scanner. The main challenge here is how to reduce the image artifacts generated from data limitations which occurs due to the scanner geometry. Within the framework of PA-MAP algorithm, a breast imaging method is developed with the help of a probabilistic atlas generated from potential normal patients. Achieved results indicate an interesting findings especially with the observation quality of microcalcifications.

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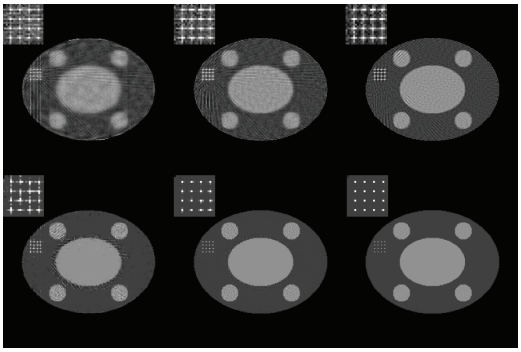

Fig. 6. Reconstruction using the proposed extended trajectory. Top are the OS-Convex reconstructions and bottom are the PA-MAP reconstructions. From left to right, the number of projection views are 8+8, 16+16 and 32+32.

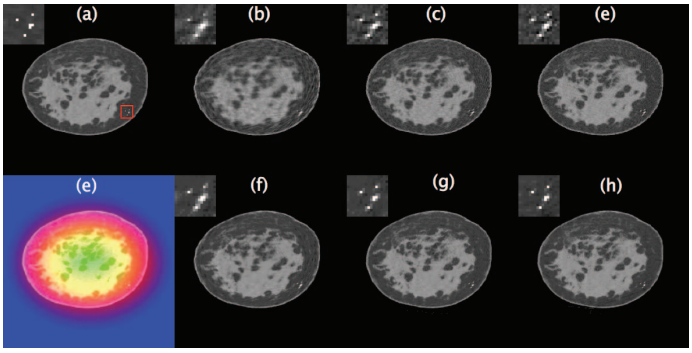

Fig. 7. (a) Breast slice with artificially added microcalcifications marked by red square. (e) is merge of breast slice registered with the probabilistic atlas. (b)-(d) are the OS-Convex reconstruction using 16+16, 32+32 and 48+48 projection views, respectively. (f)-(h) are the PA-MAP reconstruction using the same number of views of corresponding images. The magnification of the same region shown in (a) is also generated for each reconstruction for comparison.

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